965349

Course Name: **Operating Systems (BCSC 0004)**

Course Outcome

CO1: Understand the classification of operating system environment.
CO2: Understand the basic of process management.
CO3: Apply the concept of CPU process scheduling for the given scenarios.
CO4: Illustrate the process synchronization and concurrency process in operating system.
CO5: Analyze the occurrence of deadlock in operating system.
CO6: Describe and analyze the memory management and its allocation policies.
CO7: Understand the concepts of disk scheduling.

Printed Pages:4 **University Roll No. .....**

End Term Examination, Odd Semester 2024-25

BTech (CSE/ AIML & IIoT/BCT/CCV/CSF/DA/EC(CS)), II Year, III Semester

Operating Systems (BCSC 0004)

Time: 3 Hours **Maximum Marks: 50**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Consider the following set of four processes. Their arrival time and time required to complete the execution are given in the following table. All time values are in milliseconds. Calculate the average waiting time and turnaround around time using SRTF algorithm. Process | Arrival Time (T0) | Time required for completion P0 | 0 | 10 P1 | 1 | 6 P2 | 3 | 2 P3 | 5 | 4	4	3	A	P
2	Imagine you are the operating system managing a system with multiple processes. One of the processes, "Process A," is currently running but needs to be suspended because a higher-priority process, "Process B," has arrived. You need to switch to "Process B." Describe how the Process A" helps in managing the context switch to "Process B." What critical information must be stored to resume "Process A" after "Process B" completes its execution?	4	2	U	C
3	Let us consider a method used by set of two concurrent processes P1 and P2 in a uniprocessor system for accessing some shared resource. Two shared Boolean variable S1 and S2 used by process P1 and P2 to	4	4	A	P

	synchronize their activities and initial value of S1 and S2 is randomly assigned. Methods are: P1 P2 While(S1!=S2); While(S1==S2); Critical Section Critical Section S1=not S2 S1=S2 Is the method ensuring synchronization between the process P1 and P2? Justify.				
4	You are managing the operations of a server system that runs multiple tasks, each requiring exclusive access to certain resources. The system has four tasks (T1, T2, T3, T4) and four resources (R1, R2, R3, R4), each with a single instance. The resource allocation status is represented as a Resource Allocation Graph (RAG). At a particular moment: • T1 holds R1 and requests R2. • T2 holds R2 and requests R3. • T3 holds R3 and requests R4. • T4 holds R4 and requests R1. a) Draw the Resource Allocation Graph (RAG) based on the given scenario. b) a) Analyze the RAG to determine if a deadlock exists in this scenario.	4	5	C	M
5	In an automated kitchen system, multiple robotic units work around a circular table, each requiring access to two adjacent cooking stations to prepare their assigned tasks. These stations are limited in number, and only one robotic unit can use a station at any given time. If two units attempt to access the same station simultaneously, it can lead to conflicts, delays, or even a system halt. Your goal is to design a solution of this synchronization problem manage the resources efficiently, ensuring that all robotic units can complete their tasks without encountering issues like deadlock or starvation.	4	4	An	C

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	A memory system has 5 partitions of sizes: 100 KB,500 KB,200 KB,300 KB, and 600 KB. The following processes arrive in sequence with memory requirements: P1=212 KB, P2=417 KB, P3=112 KB, P4=426 KB Simulate the allocation using: a) First Fit b) Best Fit c) Worst Fit For each method, determine: i) Which processes are allocated successfully? ii) The total internal and external fragmentation.	1.5+1.5	6	U	C
7	What is thrashing in a virtual memory system? How can it be minimized?	1+2	7	R	F
8	Consider a paging hardware with a TLB. It takes 15 milliseconds to search the TLB and 90 milliseconds to access the physical memory. If the TLB hit ratio is 0.7, find the effective memory access time (in milliseconds).	3	6	A	P
9	Compare and contrast the three primary file allocation methods: Contiguous, Linked, and Indexed. Highlight their advantages and disadvantages.	3	7	U	F
10	Explain file access methods (e.g., sequential access, direct access). Write attributes of file system.	3	6	A	P

Section – C

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Consider a system with n=4 processes (P1, P2, P3, P4) and m=3 resource types (A, B, C). The total number of resources is [10,5,7]. Allocation matrix: $\begin{bmatrix} 0 & 1 & 0 \\ 2 & 0 & 0 \\ 3 & 0 & 2 \\ 2 & 1 & 1 \end{bmatrix}$ Maximum need matrix: $\begin{bmatrix} 7 & 5 & 3 \\ 3 & 2 & 2 \\ 9 & 0 & 2 \\ 4 & 3 & 3 \end{bmatrix}$ **Compute:** a) The Need matrix. b) Determine if the system is in a safe state using the Banker's Algorithm. c) If P4 requests [1,0,2], check if the request can be granted.	1+2+2	5	A	P
12	What is the difference between seek time and rotational latency time? Suppose that a disk drive has 5000 cylinders, numbered 0 to 4999. The drive currently services a request at cylinder 143, and the previous request was at cylinder 125. the queue of pending request in FIFO order is 86,1470,913,1774,948,1509,1022,1750,130 Starting from the current position, what is the total distance (in cylinders) that the disk arm moves to satisfy all pending requests, for each of the following algorithms i) SSTF ii) C-LOOK	1+2+2	7	A	P
13	You are a software engineer working on optimizing a memory management system for a critical application that runs on embedded hardware with limited RAM. The system operates with only **4-page frames** and is frequently encountering page faults, impacting performance. You have identified a representative page reference string for analysis: **Page reference string:** 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 Your task is to evaluate different page replacement strategies and recommend the best approach based on their performance.	5	6	An	P